

MATERIAL SAFETY DATA SHEET

SECTION 1. CHEMICAL IDENTIFICATION

Product name:	Fipronil 50g/L SC
Chemical Name	(±)-5-amino-1-(2, 6-dichloro- α , α , α -trifluoro- <i>p</i> -tolyl) -4-trifluoromethylsulfinylpyrazole-3-carbonitrile (IUPAC)
Synonym	None
Chemical Family	Phenylpyrazole insecticide
Chemical Formulation	C ₁₂ H ₄ Cl ₂ F ₆ N ₄ OS
Product Use:	Insecticide
Formulation:	Suspension Concentrate

SECTION 2. COMPOSITION/INFORMATION ON INGREDIENTS

Component Name	CAS No.	Content (w/v)
Fipronil	120068-37-3	50 g/l
Other ingredients	/	to 1L

SECTION 3. HAZARDS IDENTIFICATION

Acute Health Effects:

Swallowed:	Harmful if swallowed.
Eye:	Will irritate eyes. May cause weeping and redness.
Skin:	Will irritate skin. Fipronil does not readily penetrate the skin.
Inhaled:	Harmful if inhaled.

Other:

Symptoms of overexposure include respiratory difficulty, decreased reflexes, lethargy,

tremors, and in severe cases, convulsions. Repeated exposure at dose levels that are not acutely toxic may result in symptoms of toxicity.

SECTION 4. FIRST AID MEASURES

Swallowed: If swallowed, immediately contact a doctor or Poisons Information Centre, and follow the advice given. Keep under medical supervision.

Eye: Immediately irrigate with copious quantities of water for at least 15 minutes. Seek immediate medical assistance.

Skin: Remove contaminated clothing. Wash affected areas with plenty of soap and water.

Inhaled: Remove victim to fresh air. Seek medical assistance.

First Aid Facilities: Ensure washing facilities are available, including eyewash facilities.

Advice to Doctor: Fipronil is a reversible gamma-aminobutyric (GABA) receptor inhibitor. During intoxication it will induce neurological stimulation with possible convulsions. Treat symptoms. No specific antidote known. Phenobarbital, and to a lesser extent, benzodiazepines, have been shown experimentally to be effective in preventing convulsions induced by fipronil. Due to slow absorption of fipronil through the gut, symptoms of intoxication may be delayed several hours to one day. Absorption may be decreased by the use of gastric lavage, saline purgative and activated charcoal (possible entero-hepatic recirculation). Continue monitoring – due to the slow elimination of the compound.

SECTION 5. FIRE FIGHTING MEASURES

Fire/Explosion Hazard:

Extinguishing Media: Carbon dioxide, water spray, dry agent, foam.

Special Fire Fighting Procedures:

Firefighters should wear full protective gear, including self-contained breathing apparatus (AS/NZS 1715, 1716). If possible and without risk, remove intact containers from exposure to fire. Otherwise, spray unopened containers with water to keep cool. Whenever possible, contain fire-fighting water by bunding area with sand or earth to prevent it entering any bodies of water.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Spillage or leak procedures:

Avoid contact with the spilled material or contaminated surfaces. When dealing with spills do not eat, drink or smoke and wear personal protective clothing and equipment as described in PERSONAL PROTECTION section. Prevent spilled material from entering drains or watercourses. Contain spill and absorb with earth, sand, clay, or other absorbent material. Collect and store in properly labelled, sealed drums for safe disposal. Deal with all spillages immediately. If contamination of drains, streams, watercourses, etc. is unavoidable, warn the local water authority.

Neutralising Chemicals: Hydrated lime.

SECTION 7. HANDLING AND STORAGE
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Handling & Storage Precautions:

Keep out of reach of children. Store in the closed, original container in a dry, cool, well-ventilated area, out of direct sunlight. Store in a locked room or place away from children, animals, food, feedstuffs, seed and fertilizers.

SECTION 8. EXPOSURE CONTROLS/ PERSONAL PROTECTION

Engineering Controls:

Control process conditions to avoid contact. Use local exhaust ventilation during manufacture. Use in a well-ventilated area only.

Personal Protection:

Eyes: Safety goggles

Clothing: Cotton overalls buttoned to the neck and wrist (or equivalent clothing)

Gloves: Elbow-length PVC gloves

Respiratory: Do not breathe vapour or spray. Wear an organic cartridge respirator if inhalation is possible.

Other: Wash hands after use. After each day's use, wash gloves, goggles, respirator facepiece and contaminated clothing.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	White suspension
Odour:	Vegetable oil smell
Boiling Point/Melting Point:	Not determined
Vapor Pressure:	3.7×10^{-4} mpa@25°C
Specific Gravity:	1.05-1.07 at 20°C (H ₂ O=1)
Flashpoint:	Non-flammable
Flammability Limits:	Not applicable
Ignition Temperature:	410°C
Solubility in Water:	Disperses in water
pH:	4.0 – 7.0
Formulation:	Suspension Concentrate

SECTION 10. STABILITY AND REACTIVITY

Reactivity Hazards: Stable under normal conditions of use. No dangerous reaction known under normal conditions of use. Avoid extreme heat and fire. Toxic thermal decomposition products may include oxides of carbon, sulphur and nitrogen, and compounds of chlorine and fluorine.

SECTION 11. TOXICOLOGICAL INFORMATION

Acute Oral Toxicity	Rat: LD50: 1127 mg/kg.
Acute Dermal Toxicity	Rabbit: LD50: > 2493 mg/kg.
Acute Inhalation Toxicity	Rat: LC50 (4 h): > 0.5 mg/L.
Skin Irritation	Practically non-irritating to skin of rabbits.
Eye Irritation	Slightly irritating to eyes of rabbits.
Sensitization	Not a skin sensitizer to guinea pigs.

SECTION 12. ECOLOGICAL INFORMATION

These data below are for Fipronil technical:

Bird:

Acute oral LD50 for bobwhite quail 11.3, mallard ducks > 2000, pheasants 31, red-legged partridges 34, house sparrows 1120, pigeons > 2000 mg/kg. Dietary LC 50 (5 d) for bobwhite quail 49, mallard ducks > 5000 mg/kg diet.

Fish:

Acute LC50 (96 h) for bluegill sunfish 85, rainbow trout 248, European carp 430 µg/L.

Daphnia:

LC50 (48 h) 0.19 mg/L.

Algae:

EC50 (120 h) for *Selenastrum Capricornutum* > 0.16 mg/L.

Bees:

Highly toxic to honeybees both by direct contact and by ingestion. However no risk to bees when used as a soil or seed treatment.

SECTION 13. DISPOSAL CONSIDERATIONS**Waste Disposal Methods:**

Triple or preferably pressure rinse containers before disposal. Add rinsings to spray tank. Do not dispose of undiluted chemicals on site. Break, crush, or puncture and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for the purpose, clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

SECTION 14. TRANSPORT INFORMATION

The product is classified as a Dangerous Good under the Australian Code for the Transport of Dangerous Goods by Road and Rail: PESTICIDE, LIQUID, TOXIC, N.O.S. (contains fipronil), Class: 6.1, PG III, UN 2902. Must be stored and transported according to local, State/Territory and Federal regulations.

SECTION 15. REGULATORY INFORMATION**EPCRA SARA Title III Classification**

Section 311/312 Hazard Classes: Acute Health Hazard
Chronic Health Hazard

Section 313 Toxic Chemicals: None

Unbreakable packaging; put breakable packaging into closed unbreakable container.
Do not transport with food and feedstuffs.

SECTION 16. OTHER INFORMATION

Revise date: March 19, 2010
MSDS Number: SN581/50g/L SC

This information is provided in good faith but without express or implied warranty.
Buyer assumes all responsibility for safety and use not in accordance with label instructions.